

DSA 210 Project Proposal

Project Overview & Hypothesis

This project will explore how geographical separation, environmental factors, and evolving personal dynamics influence media consumption. I aim to analyze my own and my best friend's (recently turned partner's) Spotify streaming habits across one common and two different countries, covering the past 6 months – a year, thus including many personal and environmental factors.

Data Collection:

I will use a personal dataset consisting of my own and my partner's listening histories. The core data will be sourced directly from Spotify's internal user databases (Account Data). This data will be collected by submitting an "Account Data" privacy export request to Spotify for both individual accounts. The dataset will be provided in JSON format. It is a time-series dataset where each sample represents a streamed track with its detailed information: timestamp, track name, artist name and duration played. Because the export covers from 180 to 365 days of listening for two users, I expect the baseline dataset to contain well over 4000 samples.

Moreover, Spotify Web API for track-level audio metadata, and the Open-Meteo historical weather API for meteorological data will be used in order to enrich the data. Python scripts to interface with the respective APIs will be developed: <spotipy> library to query the Spotify Web API to retrieve audio features; and <requests> library to query the Open-Meteo API, passing our respective cities' geographic coordinates and specific streaming dates to pull historical weather conditions. The enrichment sets will append float values to the core data, adding valence ratings and local weather metrics — precipitation, mean temperatures.